

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 1: GST (Goods and Services Tax)

Topic-wise short notes with solved examples

GST is a single tax charged on the supply of goods and services. This chapter covers how the tax is worked out on a bill and how a dealer sets off the tax already paid.

1 What GST is

- **GST** is a tax charged on the value of goods and services each time they are supplied (sold).
- The amount on which the tax is calculated is the **taxable value**. Normally it is the selling price after any discount.
- The **rate of GST** depends on the goods or service and is always given in the question.

$$\text{GST} = \frac{\text{rate} \times \text{taxable value}}{100} \quad \text{Amount of the bill} = \text{taxable value} + \text{GST}$$

- If a discount is given: taxable value = list price – discount. GST is charged on the discounted price.

2 Sales within a state: CGST and SGST

- When the seller and the buyer are in the **same state** (intra-state supply), the GST is split into two equal parts.
- **CGST** (Central GST) goes to the Central Government and **SGST** (State GST) goes to the State Government. In a Union Territory, UTGST takes the place of SGST.

$$\text{CGST} = \text{SGST} = \frac{1}{2} \times \text{GST}$$

- So at a GST rate of 18%, CGST is 9% and SGST is 9%.

Example 2.1 An article with list price ₹8,000 is sold at a discount of 10% within the same state. If the rate of GST is 18%, find the CGST, the SGST and the amount of the bill.

SOLUTION

Discount = 10% of ₹8,000 = ₹800, so taxable value = ₹7,200

CGST = 9% of ₹7,200 = ₹648 and SGST = 9% of ₹7,200 = ₹648

Amount of the bill = 7200 + 648 + 648

CGST ₹648, SGST ₹648, bill amount ₹8,496

3 Sales between states: IGST

- When the seller and the buyer are in **different states** (inter-state supply), a single tax called **IGST** (Integrated GST) is charged at the full rate. It goes to the Central Government.

IGST = GST at the full rate

Example 3.1 A dealer in Delhi sells goods worth ₹15,000 to a dealer in Jaipur. If the rate of GST is 12%, find the tax charged and the amount of the bill.

SOLUTION

Delhi and Jaipur are in different states, so IGST is charged.

IGST = 12% of ₹15,000 = ₹1,800

IGST ₹1,800; bill amount ₹16,800

4 A bill with several items

- Work out the taxable value and the GST for each item separately, because the rates can differ, and then add.

Example 4.1 Find the amount of this bill for a sale within the same state.

Item	Price per unit (₹)	Quantity	Rate of GST
A	500	2	12%
B	1,200	1	18%

SOLUTION

Item A: taxable value = $500 \times 2 = ₹1,000$; GST = 12% of 1,000 = ₹120 (CGST ₹60, SGST ₹60)

Item B: taxable value = ₹1,200; GST = 18% of 1,200 = ₹216 (CGST ₹108, SGST ₹108)

Total taxable value = ₹2,200 and total GST = ₹336

Amount of the bill = ₹2,536

5 Finding the price before tax

- If the amount of the bill and the rate $r\%$ are known:

$$\text{Taxable value} = \frac{\text{amount of the bill} \times 100}{100 + r}$$

Example 5.1 The amount of a bill, including GST at 18%, is ₹5,310. Find the price of the article before tax.

SOLUTION

Let the taxable value be ₹ x . Then $x + \frac{18}{100}x = 5310$, so $\frac{118x}{100} = 5310$

$$x = \frac{5310 \times 100}{118}$$

The price before tax is ₹4,500.

6 Input tax credit

- A dealer pays GST when buying goods (**input GST**) and collects GST when selling them (**output GST**).
- The dealer does not pay tax twice. The tax already paid on purchases is set off against the tax collected on sales. This set-off is the **input tax credit (ITC)**.

$$\text{GST payable to the government} = \text{output GST} - \text{input GST}$$

- So each dealer effectively pays tax only on the value that he or she adds.

Example 6.1 A dealer buys an article for ₹20,000 and sells it to a consumer for ₹25,000, both within the same state. The rate of GST is 18%. Find (i) the input CGST and SGST (ii) the output CGST and SGST (iii) the net GST payable by the dealer (iv) the price paid by the consumer.

SOLUTION

- (i) Input CGST = 9% of 20,000 = ₹1,800 and input SGST = ₹1,800
(ii) Output CGST = 9% of 25,000 = ₹2,250 and output SGST = ₹2,250
(iii) Net CGST = 2250 – 1800 = ₹450 and net SGST = ₹450, so net GST = ₹900
(iv) Price paid by the consumer = 25000 + 2250 + 2250

Net GST payable ₹900 (CGST ₹450 + SGST ₹450); the consumer pays ₹29,500.

Example 6.2 A dealer in Mumbai buys goods for ₹10,000 from a dealer in Delhi and sells them for ₹12,000 to a customer in Mumbai. The rate of GST is 18%. Find the net GST payable by the Mumbai dealer.

SOLUTION

Purchase from another state: input IGST = 18% of 10,000 = ₹1,800

Sale within the state: output GST = 18% of 12,000 = ₹2,160 (CGST ₹1,080 + SGST ₹1,080)

Net GST payable = 2160 – 1800

₹360

REMEMBER

Check: the dealer added ₹2,000 of value, and 18% of ₹2,000 is ₹360.

7 Common mistakes

- Charging GST on the list price when a discount is given. Tax is on the discounted price.
- Charging both CGST and SGST at the full rate. Each is half the GST rate.
- Using CGST and SGST for a sale between two states. That sale carries IGST.
- Forgetting to subtract the input tax when the net tax payable is asked.